

Matrix functions and stability

Preservation of real H-matrix structure by Newton square-root iterates

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Rating rationale: Challenging because convergence does not control comparison-matrix structure at each iterate; specialist impact concerns square-root algorithms for H-matrices.

Lean proof and verification evidence

The complete original Newton target is Lean verified, 2026-09-17 (UTC). The [immutable proof](#) at 3312b0795873 passed [actual non-root Linux run 35175258802](#). The [retained logs and source hashes](#) and [independent runtime review](#) cover all 271 submitted inputs, all 24 Comparator targets, default-kernel replay, standard transitive axioms and the actual rejection/isolation controls.

NLA.SF01.canonical_newton_preservation proves that the original exact recurrence, initialized at $X_0 = A$, is defined and preserves the nonsingular real H-matrix property and strictly positive diagonal for every positive dimension and every iteration. The comparison matrix and full complex algebraic spectral radius match the original definitions. Spectral-to-weight and weight-to-spectral bridges are proved; no symmetry, normality, positive-entry, bounded-iteration or supplied-representation premise is added. See the [independent contracts](#) and [source correspondence](#). The manuscript's scaled/affine initializations and Halley extensions are outside this formal claim.

Formalization and verification: **George Stepaniants**, Department of Computing and Mathematical Sciences, California Institute of Technology. **Matthew J. Colbrook**, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, retains original mathematical authorship. **Sidney Holden** retains authorship and Apache-2.0 credit for two unchanged, actually imported [IV-03 source files](#). AI assistance and scoped independent agent reviews are disclosed; no official Tau Ceti or human endorsement is claimed.

The project pins Lean 4.33.1, [Mathlib](#) and [LeanCert](#). A consumed kernel-mode LeanCert certificate proves $0 < 1/2$; the dimension, matrix and iteration arguments are symbolic, with no interval subdivision. All 24 exported theorems use only *propext*, *Classical.choice* and *Quot.sound*. Run `lake build in matrix-functions-and-stability/SF-01/lean`; the [project README](#) gives the complete Linux checker commands and separates the actual local macOS development from the GitHub execution. Later publication-commit and upstream PR runs are separate checks.

The [formalization note](#) and [TeX source](#) record the scope and authorship. The original mathematical resolution remains unchanged below.

Resolution — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** **PASS for the exact target.**

Every exact Newton square-root iterate initialized at $X_0 = A$ remains a real nonsingular H-matrix with positive diagonal. The theorem includes arbitrary positive scalar scaling and nonnegative affine initializations, with one diagonal-dominance weight for all iterates; it also proves the corresponding Halley preservation result.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Theorem 1; Corollary 5 gives the Halley extension.** [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

For a real square matrix $Y = (y_{ij})$, define its comparison matrix by $M(Y)_{ii} = |y_{ii}|$ and $M(Y)_{ij} = -|y_{ij}|$ for $i \neq j$. Call Y a nonsingular H-matrix when $M(Y) = sI - B$ for some entrywise nonnegative B and real $s > \rho(B)$.

Let $n \geq 1$ and let $A \in \mathbb{R}^{n \times n}$ be a nonsingular H-matrix with $a_{ii} > 0$ for every i . Consider the exact-arithmetic recurrence

$$X_0 = A, \quad X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}A), \quad k \geq 0.$$

Is every X_k a nonsingular H-matrix with strictly positive diagonal? The known existence of these iterates and their convergence to the principal square root do not themselves establish the requested structure at every step.

References and status

N. J. Higham, *Functions of Matrices: Theory and Computation* (SIAM, 2008), §6.3, equation (6.12), §6.8.3, and Research Problem 6.25, p. 170. The author's [errata](#) correct the surrounding p. 161 theorem to real H-matrices; that restriction is adopted here. L. Lin and Z.-Y. Liu, *On the Square Root of an H-matrix with Positive Diagonal Elements*, *Annals of Operations Research* 103 (2001), 339–350, concern the square root itself. Searches on 2026-09-08 for the problem number and Newton/H-matrix structure preservation found no resolution. The recent [Bini–Iannazzo–Meini–Meng preprint](#) (20 May 2026), §§3–4, treats M-matrices, a narrower class. The latest explicit open-problem evidence found for this assertion remains the 2008 book.

Audit — 2026-09-10

Rechecked the [2026 M-matrix paper](#), §§4 and 4.2: its normalized nonsingular M-matrix class has structure-preserving Newton iterates starting from A . This is a substantive subfamily of the displayed target. Searches for Higham's Problem 6.25 and H-matrix iterate preservation found no general answer; the surviving H-matrix question still has historical evidence.